



WATERSPHERE

Installation and Operations Manual



64-EGBSP-24	24" WaterSphere with Concrete Mounting Bracket
64-EGBSP-32	32" WaterSphere with Concrete Mounting Bracket

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Section 1. SAFETY INSTRUCTIONS



Improper installation carries the potential for electrical hazards, presenting a risk of severe or even fatal injuries to pool users, installers, or bystanders due to the threat of electrical shock, and may also result in property damage.

For your safety, it is imperative that this product be installed and maintained by a licensed and qualified contractor specializing in pool equipment, adhering to local regulations. The individual responsible for maintenance should possess extensive experience in pool equipment installation and upkeep to ensure precise adherence to the instructions outlined in this manual. Prior to installation, thorough comprehension, and compliance with all provided warning notices and instructions are crucial. Failure to do so may lead to property damage, personal injury, or loss of life. Any improper installation and/or operation may void the product's warranty and pose significant electrical hazards, potentially resulting in severe harm, property damage, or fatality. It is strictly advised not to modify this equipment. Installer, please be aware that this manual contains vital information concerning the installation, operation, and safe utilization of this product, which should be communicated to the equipment's owner/operator.

Section 2. PREPARATION

This manual provides installation and operational instructions for the PAL Lighting WaterSphere.

Read the installation & operational manual before proceeding with the installation. Keep this manual for future reference.

Before starting the installation, check that you have the correct parts below. If any parts are missing, please call technical support at +1 (713)-946-4343.

2.1 PACKAGE CONTENTS

WaterSphere
1 x Sphere
1 x Adapter
6 x M4x12mm Screws

PAL Evenglow Bubbler Sold Separately

IMPORTANT NOTICE

This is a 24V light, and it requires a compatible 24V transformer for proper operation.
Refer to bullet 3.

The PAL WaterSphere is only for gunite/concrete installations. We actively discourage users from using the PAL WaterSphere on a sun shelf.

2.2 PREPARATION WORK

1. Ensure that the pool's electrical system complies with the standards outlined in the current National Electrical Code (NEC).
2. This 24V underwater light must be installed in accordance with the National Electrical Code by a certified electrician or a qualified pool technician.

NOTE: No Bonding Terminal is required on the forming shell since the fixture is an all-plastic construction and complies with NEC article 680 Requirements.

3. This underwater light is designed for use with the 64-PCR Series Drivers.

NOTE: The Junction Box and the low voltage transformer is located at least 4 (four) inches above ground level or eight (8) inches above maximum water level, whichever is higher. The Junction Box must be at least 48 inches from the edge of the pool.

CANADIAN INSTALLATION

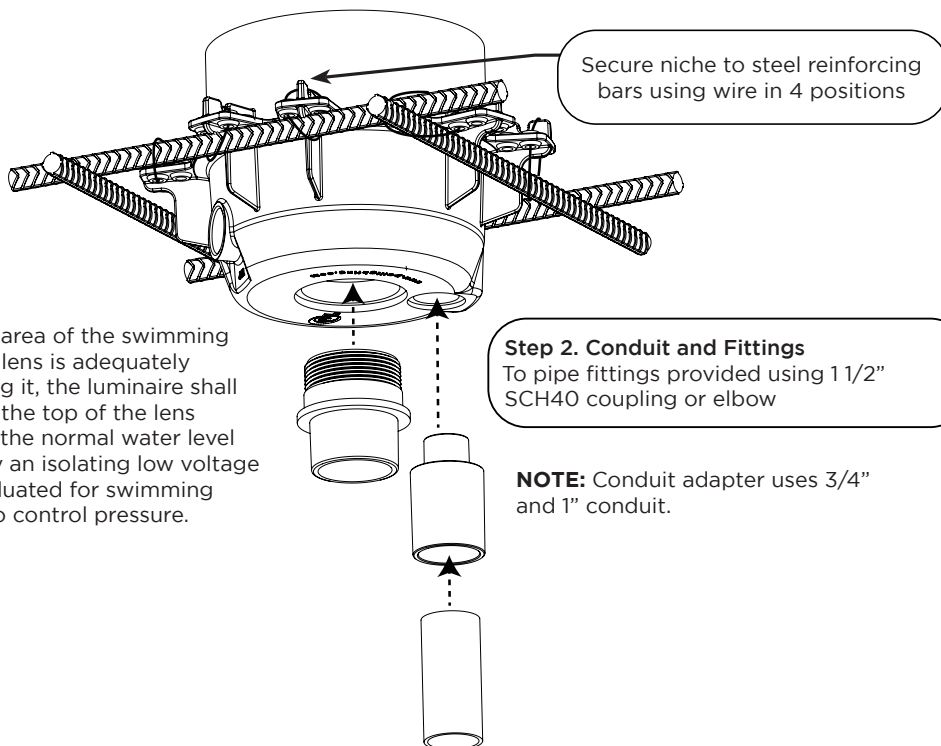
Be sure the electrical system of your pool conforms with the following requirements of the Canadian Electrical Code (CEC), CSA C22.1 and all local codes and ordinances. A licensed or certified electrician must install the electrical system to meet or exceed those requirements before the light and (fixture-housing) is installed. Failure to do so could result in death or serious injury to service technician, pool users or others due to electrical shock.

IMPORTANT NOTICE

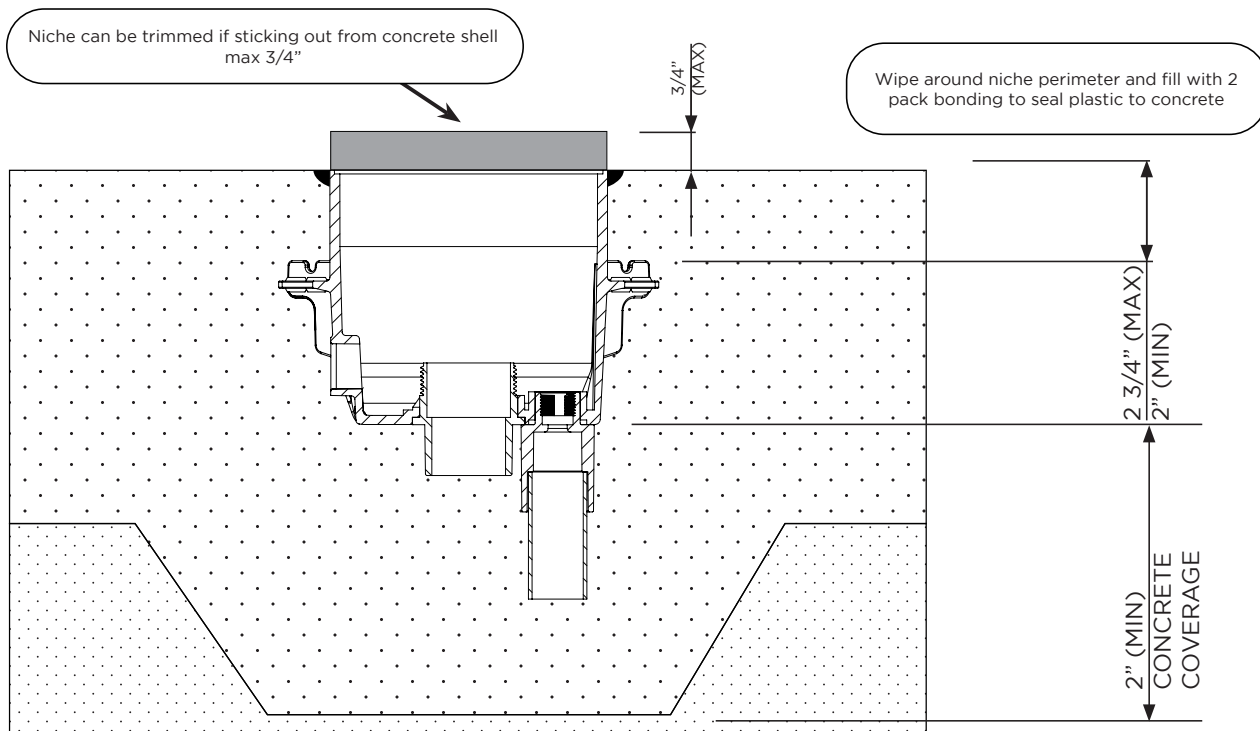
Section 3. INSTALLATION FOR GUNITE/CONCRETE POOLS

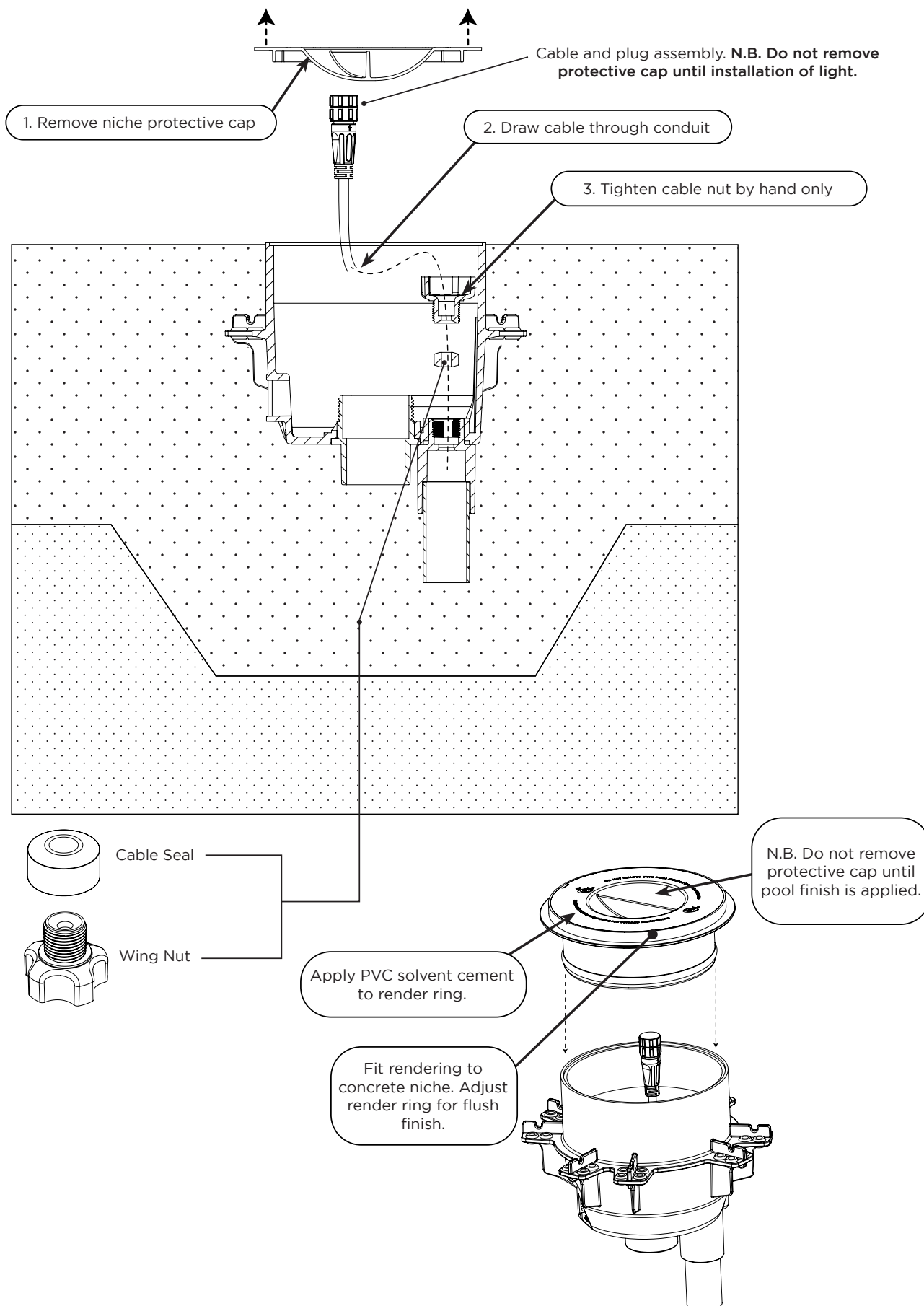
NOTE: Use PVC solvent cement on all provided fittings.

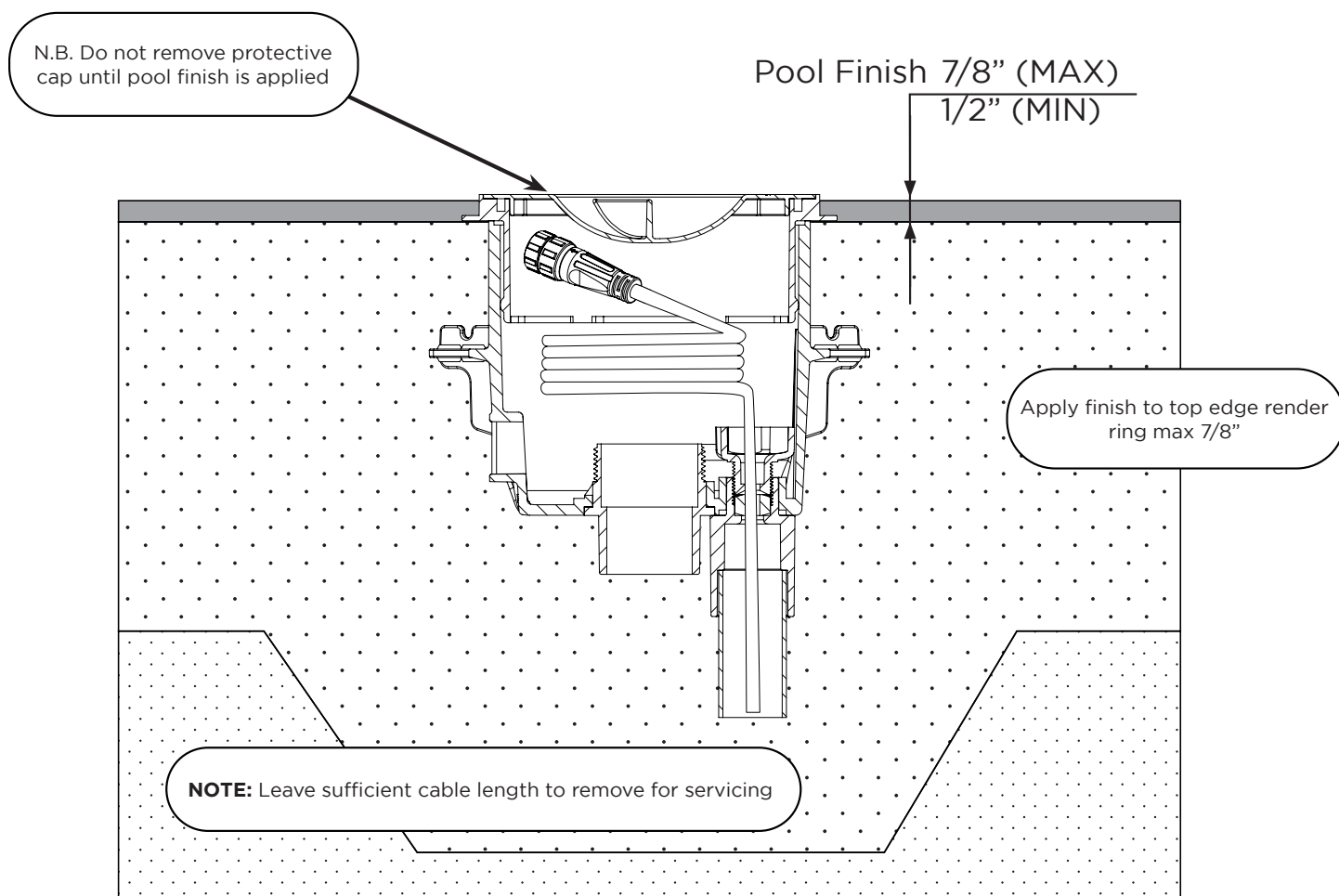
Except when the luminaire is installed in an area of the swimming pool that is not used for swimming and the lens is adequately guarded to keep any person from contacting it, the luminaire shall be installed in or on a wall of the pool, with the top of the lens opening not less than 25mm (1 inch) below the normal water level of the pool. For supply connection, use only an isolating low voltage power supply with ungrounded output, evaluated for swimming pool use. Recommended to install a valve to control pressure.



- A. Fixing Bubbler Niche to Steel
- B. Concrete Stage







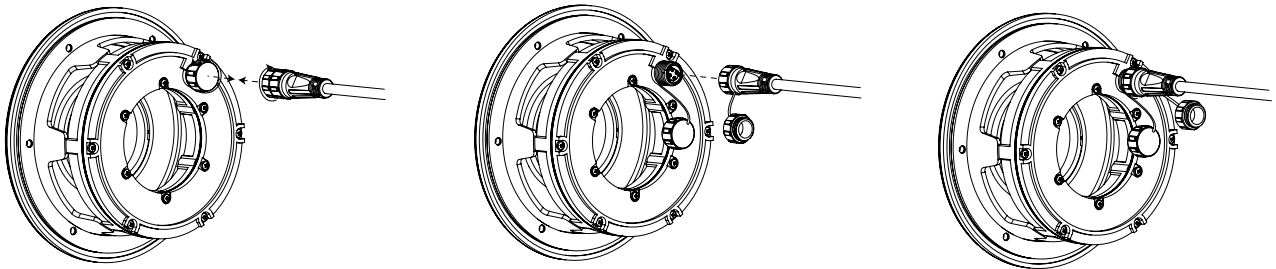
C. Installing Cable

D. Installing Luminaire

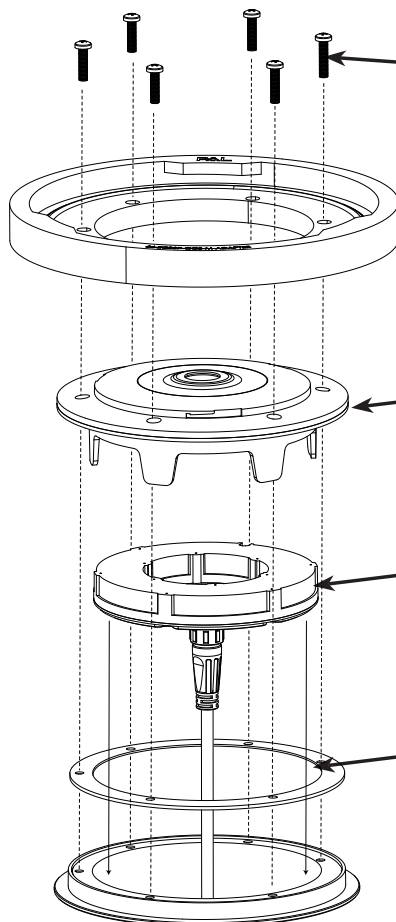
NOTE: The three machined screws will attach to the light mount and go through the gray plate into the niche to secure the light mount. The fourth pan head screw will secure only the gray gasket plate.

NOTE: The niche may cause the light mount to not align straight up and down. To fix this, turn the light mount to align with the screw hole and attach.

Remove protective caps from plug and light socket by unscrewing.



Insert plug into light and secure with screw-on cap.



Secure collar to niche using
6 x M4x12mm screws

Optional lens (only available
for 600mm and 800mm
Spheres)

Install luminaire. Luminaire rests
on ribs inside niche.

Install gasket. Gasket is placed
inside outer rib of niche.

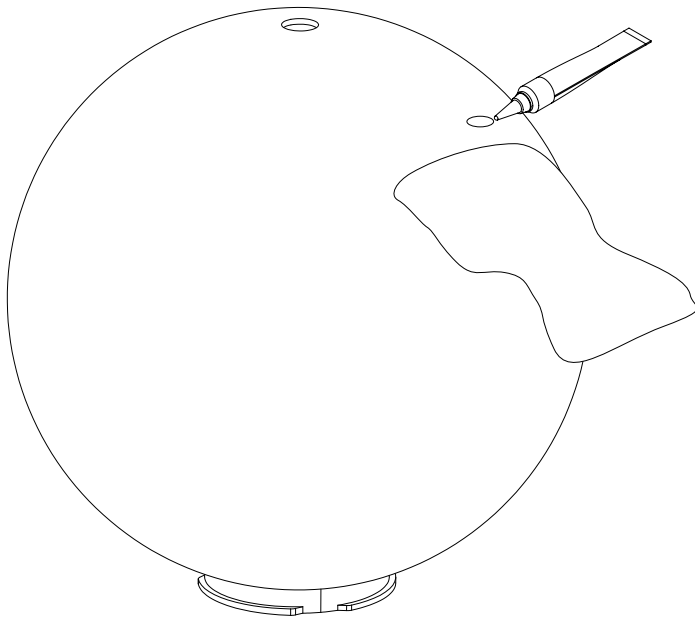
1. Once the gray face plate and light mount have been attached, pull the attached cable through the light mount and conduit back to a 24v transformer.

NOTE: When wiring into the transformer, there will be a brown and blue terminal for the wires to screw into on the left side.

2. Attach the light to the mounting plate with the thin screws at the 1, 3, and 7 o' clock positions. See Figure 4.

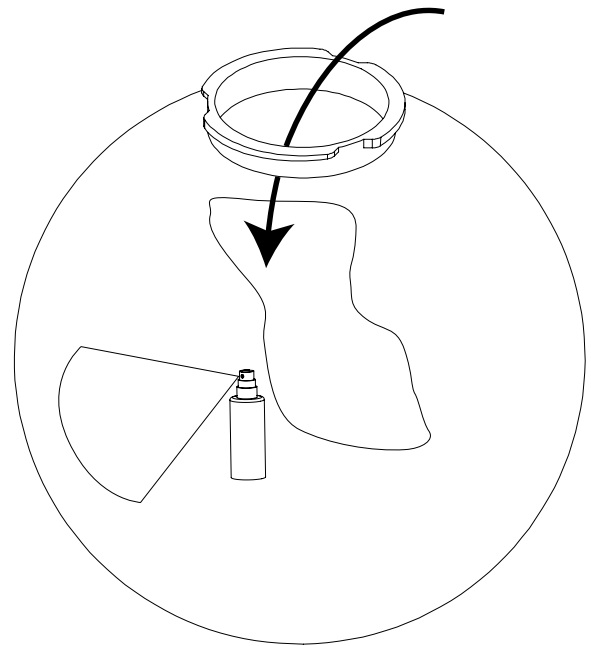
E. Sphere Preparation

Due to the nature of the production process of PAL Spheres, material thickness may not be consistent. Scratches may result from production and transport but can be easily removed using the provided paste and spray.



REMOVE SCRATCHES

Apply a pea size amount of PolyWatch over a scratch and rub using a clean cloth until scratch is removed.

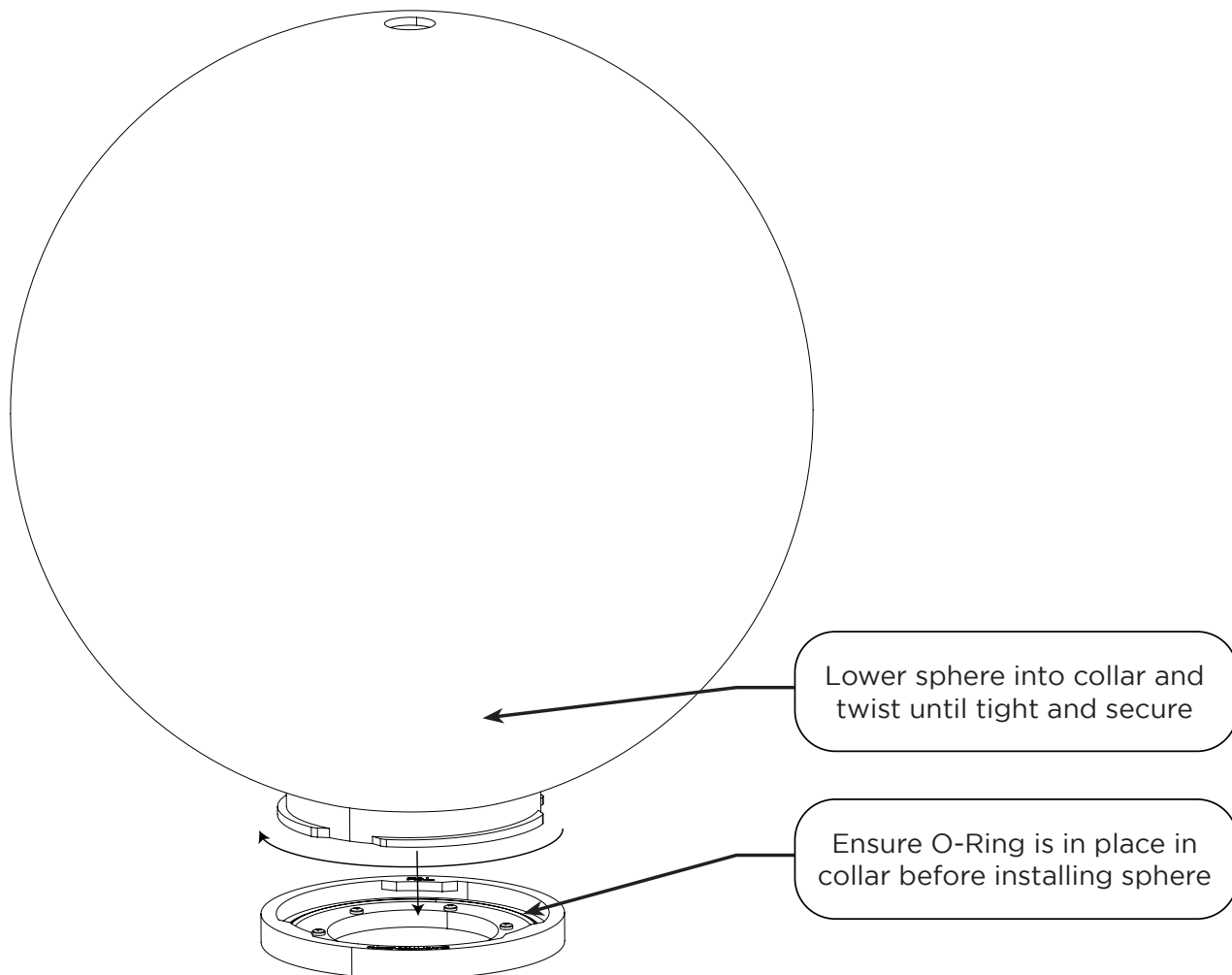


REMOVE HAZE

On inner-surface of Sphere, spray Anti-Fog Spray and polish with a clean cloth.

F. Fixing Water Sphere to the Bubbler

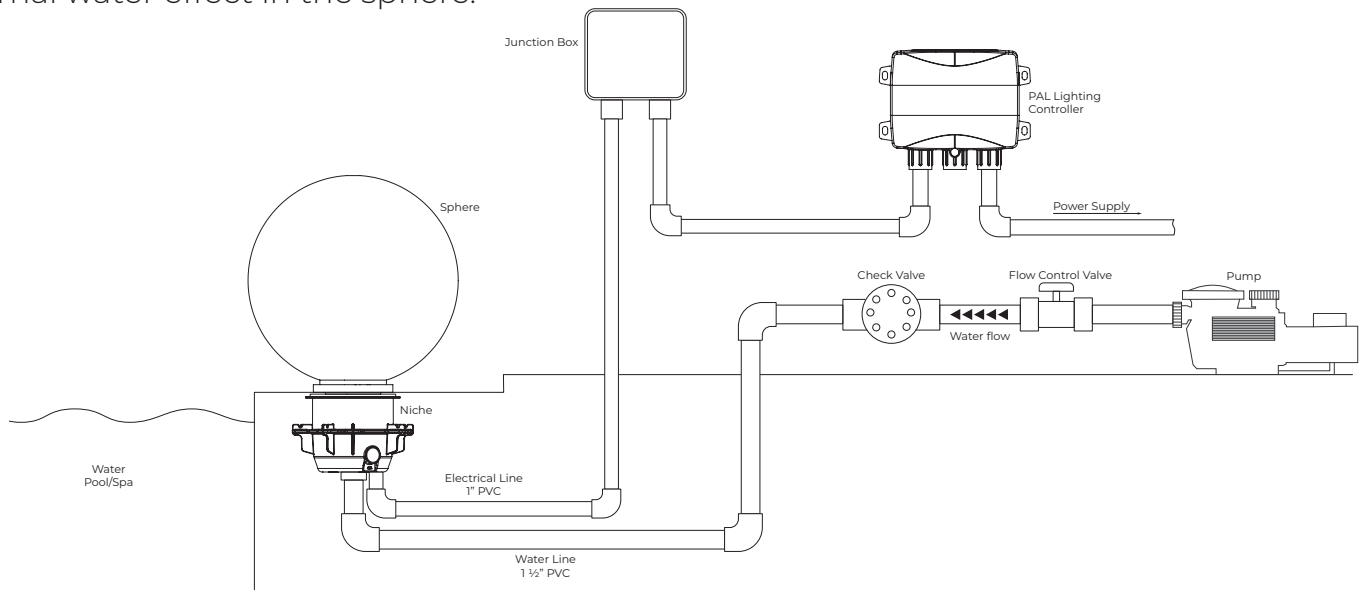
1. Attach the gray sphere adapter to the bubbler collar: Align the hole patterns on the adapter with those on the collar.
2. Screw the adapter in securely.
3. Attach the sphere to the adapter: Insert the sphere into the adapter and rotate to lock it in place using the bayonet fitting mechanism.



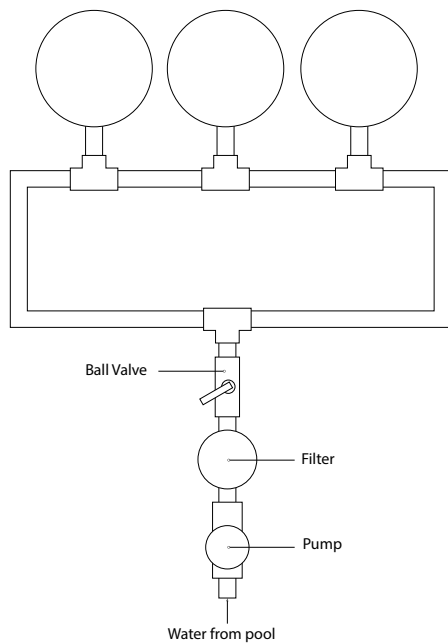
Overview

Refer to the diagram on the below for a basic installation overview.

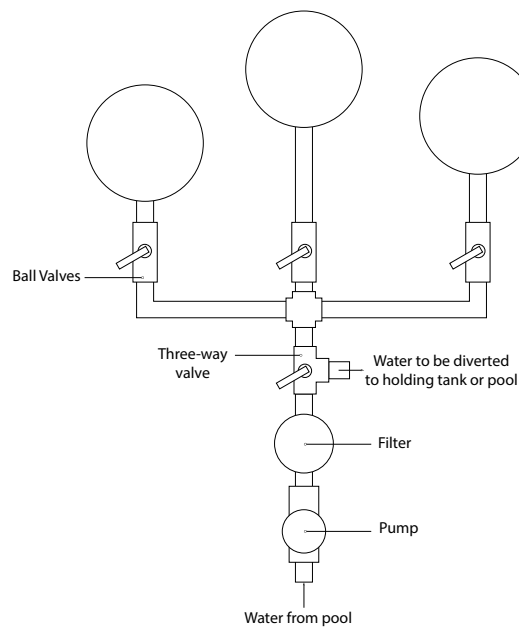
Note: Plumbing for the WaterSphere requires a check valve to prevent drain off when the system is not in use. A flow control valve should also be used in order to obtain the optimal water effect in the sphere.



**Spheres at equal distances
Use looped plumbing**



**Spheres at Different Distances
Use individual flow control valves**

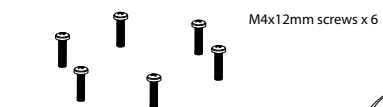
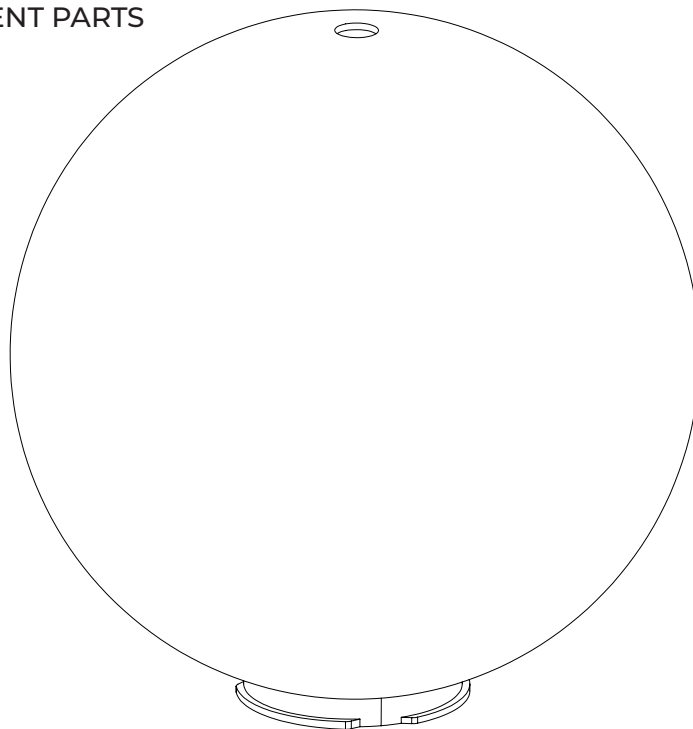


Pressure of water exiting pump to watersphere is between 30-34 GPM / 113-128 LPM. Calculate the required RPM based on your pump's specifications.

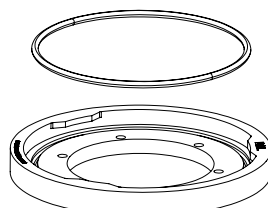
Note: When you are plumbing more than one WaterSphere, add the total volume of both WaterSphere together to determine the flow rate required.

WARNING** The Pumps automatic priming should be disengaged . Self-priming could result in personal injury or damage to the WaterSphere.

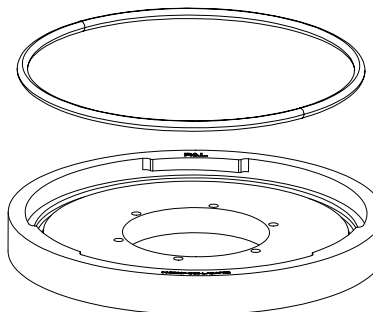
SECTION 4 - REPLACEMENT PARTS



M4x12mm screws x 6



Medium Adapter
600mm (24") Sphere



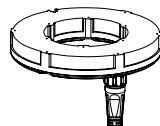
Large Adapter
800mm (32") Sphere



Lens Adapter Optional

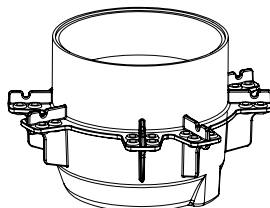


Must use silicone
gasket otherwise
screw will be too long



PAL Evenglow Bubbler
(Sold Separately)

Part Number
64-EGB-CGS-XXX



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